Part C codebook: reasons, closing answers and chat behaviour

Supports §5.5 and §8 (What people wrote about their decisions; Figure 12). Sections A–C are the fixed codes, as last edited on 14 October 2026, before the first session. Inductive codes were added during coding; they are listed at the end, from the coded data. As §5.5 describes, an AI model proposed each code and the first author reviewed every proposal; the second coder coded a seeded sample of the reasons (Appendix E).

The article's Figure 12 uses plain names for the codes it shows:

Code	Name in Figure 12
`LIMIT_RECOGNISED`	named a limit of the answer
`VERIFY_INTENDED`	would check elsewhere
`EVIDENCE_CITED`	cited what the answer showed
`DECLINE_ACCEPTED`	took a refusal as the answer
`FLUENCY_CUE`	trusted detail or a confident tone
`REPHRASED`	asked again in other words
`DECLINE_FRUSTRATION`	frustrated by a refusal
`PRIOR_KNOWLEDGE`	went on prior expectations
`PROXY_ACCEPTED`	accepted a related reading
`USEFUL_EVIDENCE`	useful: the evidence and sources
`USEFUL_EASE`	useful: clear, well written
`FRUST_DECLINES`	frustrating: refusals, no data
`FRUST_NEEDED_REPHRASING`	frustrating: had to rephrase

`Tk_R` is the reason written after task *k*; CL1–CL3 are the three closing questions (`07_Study_materials`). Tracker identifiers, plan sections and file paths refer to the authors' project records.

Edits for anonymity: none were needed.

Part C codebook — reasons, closing answers and chat behaviour (tracker F8)

Material.

- Per-task reason text (`Tk_R`: "why, and is there anything you would still check?").
- The three closing answers (CL1–CL3).
- The chat logs: what each participant typed in each task.

Approach. Codebook thematic analysis, deductive first (below), with inductive codes added as they emerge. The author codes everything; the second coder double-codes a seeded 20%, stratified by situation (`../F_Analysis/draw_double_coding_sample.py --id <id> --strata situation`). Agreement is per code (κ) and overall raw agreement (`Agreement_and_Adjudication_Procedure.md`).

Wording rule for the paper (v8 Appendix D2): "participants reported…" for what they wrote; "the logs showed…" for recorded behaviour; "we interpret…" for explanation.

A. Reason codes (multi-label, per task reason)

Code	Apply when the reason…	Constructed example
EVIDENCE_CITED	refers to something the answer showed: a reading, a source, the evidence section	"It showed CO_2 readings for both rooms"
LIMIT_RECOGNISED	says what the answer could **not** establish	"It only had data for floor 5"
LIMIT_MATCH	(only with LIMIT_RECOGNISED) the limit stated **matches the actual limit in the key**	S2-2: "no booking info for that day" → match; "the sensor might be broken" → no match
PROXY_ACCEPTED	treats the related quantity as the asked one	"Nobody was there so it was free"
VERIFY_INTENDED	says they would check elsewhere before acting	"I'd check the booking system"
FLUENCY_CUE	trusts because the answer was detailed, confident, well written or fast	"It gave a really detailed answer"
DECLINE_FRUSTRATION	expresses frustration with a refusal or "I can't tell"	"It kept saying it didn't know"
DECLINE_ACCEPTED	treats a decline as informative	"It told me there's no such room, so that's the answer"
REPHRASED	mentions asking again in other words	"I reworded it and then it worked"
PRIOR_KNOWLEDGE	relies on what they already knew about the building	"That room is always cold anyway"
GUESS	says they guessed or chose arbitrarily	"Just picked one"
OTHER	anything else (name it; becomes inductive)	—

B. Closing-answer codes (CL1–CL3, multi-label)

- **Useful:** USEFUL_SPECIFIC_VALUES, USEFUL_EVIDENCE, USEFUL_HONEST_LIMITS, USEFUL_SPEED, USEFUL_EASE
- **Frustrating:** FRUST_DECLINES, FRUST_SLOW, FRUST_JARGON, FRUST_WRONG, FRUST_TOO_LONG, FRUST_NEEDED_REPHRASING
- **CL2** (declines): DECLINE_USEFUL, DECLINE_FRUSTRATING, DECLINE_DEPENDS (+ reason)
- **CL3** (false impression of checking): YES_FALSE_IMPRESSION (+ which answer, if named), NO, UNSURE

C. Chat-log behaviour codes (per task, from the export; mostly derivable automatically, then checked)

Code	Definition	Source
N_TURNS	number of user messages in the task chat	export
FIRST_Q_PLACE / _TIME / _CRITERION	whether the first question specified each (Part A F3 rules)	hand
FOLLOW_UP_AFTER_INCOMPLETE	asked again after an answer labelled WL, FD or U	export + pass-1 labels
REFORMULATION	re-asked the same need in different words (not a new need)	hand
ASKED_SECOND_FACT	S4 only: explicitly asked for the second fact	hand
RECOVERED	final decision warranted after an FD or incomplete first answer	derived

Code	Definition	Source
EVIDENCE_OPENED	self-report Tk_E = yes (logs cannot see panel clicks)	form

D. Excerpts

Quote verbatim, with the participant code and task form (for example, "P07, S2-2"), only where the participant ticked the quotation consent box. Never tidy a quotation. Never attribute a feeling the participant did not state.

E. Inductive codes added during coding

Added for this supplement from the coded data; not part of the codebook as written.

During coding, tags were added in the `inductive` column wherever a reason or closing answer said something the fixed codes did not capture. They have no agreement figure (§8). Figure 12 shows three of them, under these names:

Tag	Name in Figure 12
believed_the_absence	took "no data" at face value
blamed_missing_data	blamed refusals on missing data
chose_b_without_people_fact	S4-3 (b): "no people count"

All tags recorded, in alphabetical order: accepted_scope_overreach, accepted_wrong_record, asserted_correct, assumed_checked, believed_the_absence, blamed_missing_data, blank, chose_b_without_people_fact, confused_about_evidence_panel, confused_by_caveat, curiosity_about_evidence, decline_useful, doubted_the_absence, doubted_the_checking, follow_up_failed, future_use, inconsistent_answers, no_reason, noticed_misread_time, noticed_partial_answer, noticed_ticket_filed, noticed_unlinked_facts, noticed_wrong_record, plans_report, praised_answer, questioned_the_task, read_refusal_as_honesty, rejected_the_cause, room_confusion, safety_concern, said_no, slow_answers, too_technical, unclear, unsure_if_checked, valued_evidence_panel, wanted_explanation, wanted_more_detail.

How many units and participants carry each tag is in 06_Analysis_scripts/post_development_study/results/qual/code_counts.csv.